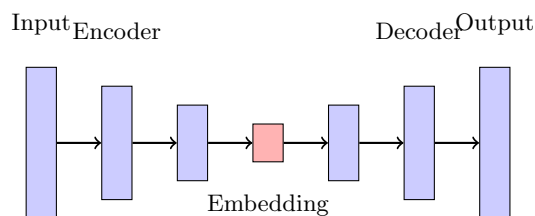


1 Encoder/Decoder Architecture

- An encoder/decoder architecture performs downsampling to a dense representation (encoding), then upsamples to predict the next token (decoding).
- Assumes that large, high/dimensional data can be represented by a lower dimensional, dense vector.
- The dense vector is also called an **embedding vector**.
 - For GPT, the embedding vector is $\mathcal{R}^{768}$

1.1 Encoder/Decoder Funnel Structure



1.2 Use Cases for Embedding Vectors

- Finding pages on the internet that are similar
 - Convert each item to an embedding vector, compute degree of similarity with cosine similarity scores, edit distance, etc.
- Compute new embedding vectors as weighted average of other embedding vectors

2 Preventing Model Hallucinations

There are several ways of preventing model hallucination:

- Retrieval Augmented Generation (RAG): Model uses embedding vector to query database for precomputed embeddings that are accurate